Zihao Zhang

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https://zizizi-hao.github.io/ | https://github.com/Zizizi-hao

RESEARCH INTERESTS

3D/4D world model for Physical AI (Embodied AI, Autonomous Driving)

EDUCATION

Shenzhen Loop Area Institute (SLAI) Tutor: Ping Luo and Hongyang Li Sept 2025 – present

- Coursework: Geometry and Learning for 3D Vision (A-), Machine Learning System (A-), Natural Language Processing and Large Language Models (A)

Southeast University (SEU) Tutor: Ziyuan Pu Sept 2024 – present

- Coursework: Optimization, Game Theory and Learning (94/100)

Hunan University (HNU) Tutor: Chenhui Liu Sept 2021 – June 2024

- Coursework: Traffic Simulation and Transportation Big Data (92/100)

Hefei University of Technology (HFUT) Tutor: Huanping Pang Sept 2017 – June 2021

- Coursework: Traffic Planning and Transportation Management (90/100), Object-Oriented Programming(90/100),

RESEARCH EXPERIENCE

Dexterous Eye-Hand Coordination System Development and Application, Participation Oct 2025 – present

- Led the deployment of a dual-arm dexterous manipulation demo at SLAI, successfully delivering 10+ demonstrations at major events.

Research on Trusted Federated Learning Method for Collaborative Perception of Urban Open Roads, Leader Mar 2025 – Mar 2026

- Leveraging trustworthy AI principles, I design robust and interpretable vehicle-infrastructure cooperative perception algorithms for intelligent transportation systems.

Research and Application of Key Technologies for Intelligent Driving Interactive Simulation Testing in Complex Traffic Environments of Urban Roads, Participation Sept 2024 – Sept 2025

- We establish a vehicle-infrastructure cooperative perception testbed on urban open roads, with Shuanglong Avenue near Southeast University as the primary experimental site.

Traffic Safety Analysis for Food Delivery Riders with Naturalistic Cycling Data, Participation Sept 2021 – Apr 2024

- We collected natural riding and questionnaire data from 46 food delivery riders, integrated 100,000 km of trajectory data with weather and road conditions, detected dangerous behaviors including speeding and wrong-way riding, and built an interactive visualization platform with R Shiny.

PUBLICATIONS

Peer-Reviewed Conference Paper

- **Zhang, Z.**, Liu P., Pu Z. (2025) Risk-Informed Diffusion Transformer for Long-Tail Trajectory Prediction in the Crash Scenario, TRB 104th Annual Meeting, January 5-9, 2025, In Washington, DC, U.S.. [Paper]
- Liu P., **Zhang, Z.**, Pu Z., et al., (2025) Roadside Monocular 3D Detection via Cross-View Semantic Alignment , 28th IEEE International Conference on Intelligent Transportation Systems(ITSC), November 18-21, 2025, In Gold Coast, Australia. [Paper]
- **Zhang, Z.**, Liu C., (2023) Why the Elderly Do Not Travel: An Analysis Based on the 2017 NHTS Data, CICTP2023, July 14–17, 2023, In Beijing, China.[Paper]

Peer-Reviewed Journals

- Zeng, B., Yu, X., **Zhang, Z.** and Pu, Z. (2025). Optimizing charging station locations for mixed traffic of electric and gasoline vehicles: a stochastic user equilibrium approach. Transportation Research Part E: Logistics and Transportation Review, 204, 104438. [Paper]
- **Zhang, Z.**, Liu, C., (2024). Exploration of the Heterogeneity among Elderly Drivers by Analyzing Traffic Crash Data: A Case Study in Pennsylvania, USA. SAE International Journal of Transportation Safety, 12 (3), 1–9. [Paper]
- **Zhang, Z.**, Liu, C., (2024). Identification of the factors influencing speeding behaviour of food delivery e-bikers in China with the naturalistic cycling data. International Journal of occupational safety and ergonomics.30(4):1288-1297. [Paper]
- **Zhang, Z.**, Liu, C., (2023). Exploration of riding behaviors of food delivery riders: A naturalistic cycling study in Changsha, China, Sustainability 15.23 (2023): 16227. [Paper]

Preprint & Under Review

- **Zhang Z.**, Tian H., Li T., Pu Z. and Yang Z. (2026). Do Better Visual Representations Always Lead to Better End-to-End Autonomous Driving? [Preprint] [Paper]
- Jing C., Tian H., Li H., **Zhang Z.** and Yang Z. (2026). SimVTA: Simulation-Grounded Video-Trajectory Alignment for Driving World Action Models. [Preprint] [Paper]
- **Zhang Z.**, Zhang X., Zeng B. and Pu Z. (2026). Reinforcing End-to-End Autonomous Driving Safety via Negative-Aware Planning in a Unified WAM. [Preprint] [Paper]
- He J., Zhao X., **Zhang Z.**, Tang Y. and Zhu X. (2026). CoWM-Forecast: A Collaborative Occupancy World Model for Dynamic 4D Scene Forecasting. [AAAI 2027, under review]

COMPETITIONS

The 25th China Robotics and Artificial Intelligence Competition - Autonomous Driving Simulation Special Competition Aug 2023

- Set up a simulation environment on a local Linux system based on the Baidu Apollo platform and learn the relevant C++ source code.
- I have completed the tuning of autonomous driving planning algorithms for a total of 10 scenarios, and won the second prize with a score of 400 (out of 500) in the final competition.

"Chang'an Automobile Cup" Global Chinese College Students Data Application Innovation Competition Dec 2022

- Based on data from 113 different scenarios of autonomous driving vehicles, classify the scenarios and identify dangerous driving situations for autonomous vehicles.
- Based on indicators such as the reciprocal of collision time, complete scene classification using a decision tree model, and implement danger warning using the LSTM algorithm.

AWARDS

- Hunan University Outstanding Graduates, 2024
- First Prize Scholarship, Hunan University, 2022, 2023
- "Merit Student" Awarded, Hefei University of Technology, 2020

OTHERS

Languages: Python, C++

Software: Cursor, Claud code

Reviewer: ECCV, NeurIPS, and AAAI, contributing to the autonomous driving research community.

Community Contributions: Core contributor to UniAD 2.0, updating its source code and model weights; **the repository has 4.7k GitHub stars.**